

**[BUG] Jacobian with diff\_method='best' and 'adjoint' mismatches for default**

<https://github.com/PennyLaneAI/pennylane/issues/2273>

ROW 117

# [BUG] Jacobian with diff\_method='best' and 'adjoint' mismatches for default.qubit #2273

New issue

Closed

#2274



chaeyeunpark opened on Mar 3, 2022 · edited by chaeyeunpark

Edits ...

## Expected behavior

The following code should work

## Actual behavior

Raises an error

## Additional information

No response

## Source code

```
import pennylane as qml
import pennylane.numpy as np
import pytest
from scipy.stats import unitary_group

def circuit_ansatz(params, wires):
    """Circuit ansatz containing all the parametrized gates"""
    qml.QubitStateVector(unitary_group.rvs(2**4, random_state=0)[0], wires=wires)
    qml.CRX(params[0], wires=[wires[3], wires[0]])

def test():
    dev_def1 = qml.device("default.qubit", wires=range(4))
    dev_def2 = qml.device("default.qubit", wires=range(4))

    def circuit(params):
        circuit_ansatz(params, wires=range(4))
        return qml.expval(qml.PauliX(0)), qml.expval(qml.PauliY(1))

    params = np.array([1.0])

    qnode_def = qml.QNode(circuit, dev_def1)
    qnode_adj = qml.QNode(circuit, dev_def2, diff_method="adjoint")

    j_def = qml.jacobian(qnode_def)(params)
    j_adj = qml.jacobian(qnode_adj)(params)

    print(j_def - j_adj)

    assert np.allclose(j_def, j_adj)

if __name__ == '__main__':
    test()
```

## Tracebacks

No response

## System information

```
Name: PennyLane
Version: 0.22.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /home/chaeyeun/work/Xanadu/pennylane
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, retworkx, scipy, semar
Required-by: PennyLane-Lightning, PennyLane-Qchem
Platform info: Linux-5.13.0-1029-oem-x86_64-with-glibc2.29
Python version: 3.8.10
Numpy version: 1.21.0
Scipy version: 1.7.0
```

## Assignees

No one assigned

## Labels

bug

## Type

No type

## Projects

No projects

## Milestone

No milestone

## Relationships

None yet

## Development

Fixes the operation derivative  
PennyLaneAI/pennylane

## Notifications

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## Participants



Visual

```
Platform info:      Linux-5.13.0-1024-oem-x86_64-with-glibc2.29
Python version:     3.8.10
Numpy version:      1.21.0
Scipy version:      1.7.0
Installed devices:
- lightning.qubit (PennyLane-Lightning-0.22.0.dev13)
- default.gaussian (PennyLane-0.22.0.dev0)
- default.mixed (PennyLane-0.22.0.dev0)
- default.qubit (PennyLane-0.22.0.dev0)
- default.qubit.autograd (PennyLane-0.22.0.dev0)
- default.qubit.jax (PennyLane-0.22.0.dev0)
- default.qubit.tf (PennyLane-0.22.0.dev0)
- default.qubit.torch (PennyLane-0.22.0.dev0)
```

## Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



**chaeyeunpark** added **bug** on Mar 3, 2022

**chaeyeunpark** changed the title **[BUG] Jacobian with diff\_method='parameter-shift' and 'adjoint' mismatches for default.qubit** **[BUG] Jacobian with diff\_method='best' and 'adjoint' mismatches for default.qubit** on Mar 3, 2022



**josh146** on Mar 3, 2022 · edited by josh146

Edits ▾ Member ...

Interesting 🤔 It looks like the adjoint might be incorrect here, if you also compare parameter-shift, it agrees with backprop.

Edit: yep, the adjoint method for `default.qubit` is returning the incorrect result.



**josh146** on Mar 3, 2022

Member ...

Also it doesn't appear in v0.21, so this is a bug in master!



**chaeyeunpark** on Mar 3, 2022 · edited by chaeyeunpark

Edits ▾ Contributor Author ...

@**josh146** Thanks Josh. This code is in fact from `Lightning`'s tests, which worked well yesterday. A recent commit [#2256](#) must have entered this bug.



**josh146** on Mar 3, 2022 · edited by josh146

Edits ▾ Member ...

@**chaeyeunpark** the issue seems to be with the `qml.matrix()` function. E.g., consider

```
>>> op = qml.CRX(0.5, wires=[3, 0])
>>> H = qml.generator(op, format="observable")
>>> H.ops[0]
Projector([1], wires=[3]) @ PauliX(wires=[0])
>>> H.ops[0].get_matrix()
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 1.]
 [0. 0. 1. 0.]]
>>> qml.matrix(H.ops[0])
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 1.]
 [0. 0. 1. 0.]]
>>> qml.matrix(H)
[[ 0. +0.j  0. +0.j  0. +0.j  0. +0.j]
 [ 0. +0.j  0. +0.j  0. +0.j -0. +0.j]
 [ 0. +0.j  0. +0.j  0. +0.j  0. +0.j]
 [ 0. +0.j  0. +0.j  0. +0.j  0. +0.j]]
```





chaeyeunpark on Mar 3, 2022 · edited by chaeyeunpark

Edits Contributor Author ...

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josh146 on Mar 3, 2022 · edited by josh146

Edits Member ...

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[[ 0. +0.j  0. +0.j  0. +0.j  0. +0.j]
 [ 0. +0.j  0. +0.j  0. +0.j -0.5+0.j]
 [ 0. +0.j  0. +0.j  0. +0.j  0. +0.j]
 [ 0. +0.j -0.5+0.j  0. +0.j  0. +0.j]]
```



For some reason, `qml.matrix(H)` is sorting by wires when it shouldn't be



josh146 linked a pull request that will close this issue [Fixes the operation derivative #2274](#) on Mar 3, 2022



josh146 on Mar 3, 2022

Member ...

Hey @chaeyeunpark, could you check if [#2273](#) fixes the issue on the lightning tests? The branch name is `fix-adjoint`.



chaeyeunpark on Mar 3, 2022

Contributor Author ...

Hi @josh146, thanks for your agile response. Yes, that fixes the issue.



josh146 closed this as [completed](#) in [#2274](#) on Mar 4, 2022

# Fixes the operation derivative #2274

<> Code

Merged josh146 merged 5 commits into master from fix-adjoint on Mar 4, 2022

Conversation 4 Commits 5 Checks 0 Files changed 2 +22 -1

 josh146 commented on Mar 3, 2022 Member

**Context:** With the new `qml.matrix()` function, the wire ordering was not being correctly passed when computing the matrix of the generator.

**Description of the Change:** Passes the wire ordering.

**Benefits:** Fixes a bug where the incorrect permutation of the generator matrix was being returned.

**Possible Drawbacks:** n/a

**Related GitHub Issues:** n/a

fix adjoint 62bed38

 josh146 linked an issue on Mar 3, 2022 that may be closed by this pull request

**[BUG] Jacobian with diff\_method='best' and 'adjoint' mismatches for default.qubit #2273** Closed

1 task

 github-actions bot commented on Mar 3, 2022 Contributor

Hello. You may have forgotten to update the changelog!  
Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.

Reviewers

chaeyeunpark ✓

Assignees

No one assigned

Labels

bug

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

☒ [BUG] Jacobian with diff\_method='best' and...

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2 participants

 